



## 4.2.5 Why using the Method of Normal Equations could be bad



4.2.5 Amplification of error by the Method of Normal Equations

fit width

**Homework 4.2.5.1.** Show that  $\kappa_2(A^H A) = (\kappa_2(A))^2$ .

[▼ Hint](#)   [▼ Solution](#)

Let  $A = U \Sigma V^H$  be the reduced SVD of  $A$ . Then

$$\begin{aligned} \kappa_2(A^H A) &= \|A^H A\|_2 \|(A^H A)^{-1}\|_2 \\ &= \|(U \Sigma V^H)^H U \Sigma V^H\|_2 \|((U \Sigma V^H)^H U \Sigma V^H)^{-1}\|_2 \\ &= \|V \Sigma^2 V^H\|_2 \|V (\Sigma^{-1})^2 V^H\|_2 \\ &= \|\Sigma^2\|_2 \|(\Sigma^{-1})^2\|_2 \\ &= \frac{\sigma_n^2}{\sigma_{n-1}^2} = \left(\frac{\sigma_n}{\sigma_{n-1}}\right)^2 = \kappa_2(A)^2. \end{aligned}$$

Use the SVD of  $A$ .

Let  $A \in \mathbb{C}^{m \times n}$  have linearly independent columns. If one uses the Method of Normal Equations to solve the linear least squares problem  $\min_x \|b - Ax\|_2$  via the steps

- Compute  $B = A^H A$ .
- Compute  $y = A^H b$ .
- Solve  $B\hat{x} = y$ .

the condition number of  $B$  equals the square of the condition number of  $A$ . So, while the sensitivity of the LLS problem is captured by

$$\frac{\|\delta \hat{x}\|_2}{\|\hat{x}\|_2} \leq \frac{1}{\cos(\theta)} \kappa_2(A) \frac{\|\delta b\|_2}{\|b\|_2}.$$

the sensitivity of computing  $\hat{x}$  from  $B\hat{x} = y$  is captured by

$$\frac{\|\delta \hat{x}\|_2}{\|\hat{x}\|_2} \leq \kappa_2(A)^2 \frac{\|\delta y\|_2}{\|y\|_2}.$$

If  $\kappa_2(A)$  is relatively small (meaning that  $A$  is not close to a matrix with linearly dependent columns), then this may not be a problem. But if the columns of  $A$  are nearly linearly dependent, or high accuracy is desired, alternatives to the Method of Normal Equations should be employed.

**Remark 4.2.5.1.** It is important to realize that this squaring of the condition number is an artifact of the chosen algorithm rather than an inherent sensitivity to change of the problem.